Exam Prep AI - Mock Question Paper

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

First Semester B.Tech Degree Regular Examination December 2024 (2024 Scheme)

Reg No.:_______________

Name:__________________________

03UCEST105122402

Course Code: UCEST105

Course Name: ALGORITHMIC THINKING WITH PYTHON

Max. Marks: 60

Duration: 2 hours 30 minutes

PART A

(Answer all questions. Each question carries 3 marks)

CO Marks

1

You are given a list of numbers in the format [a, b, c] where a, b, and c are unique positive integers. Write a Python program to find the missing number in the list.

CO-1

(3)

2

Draw a flowchart to find the first 'n' numbers in the Fibonacci sequence.

CO-1

(3)

3

Explain the concept of greedy algorithm and provide an example of its application.

CO-2

(3)

4

Write a pseudocode to implement a program to find the maximum value in a list of numbers using a for loop.

CO-2

(3)

5

Write the output produced by the following code.

(i) for count in range(10,20,2):

print(count, end=" ")

(ii) for i in range(10,-1,-2):

print(i)

(iii) 3*4**2**2

CO-3

(3)

6

Write a Python program to find the sum of odd numbers from N given numbers.

CO-3

(3)

7

What is the motivation for using a randomized approach in problem-solving?

CO-4

(3)

8

You are a software engineer working on a security application for a company that handles sensitive user data. The application has a 5-digit alphanumeric password system for authentication. However, due to a system error, a user has forgotten their password. Write an algorithm for recovering the 5-digit alphanumeric password.

PART B

(Answer any one full question from each module, each question carries 9 marks)

Module -1

9

a)

Explain how backtracking strategy can be applied to solve N-Queens problem?

CO-1

(5)

b)

Your college is located in a metropolitan city and you are new to that city. You would like to go out for dinner with your friends. Explain how heuristics approach can be used to find the best restaurant for dinner.

CO-1

(4)

10

a)

You are asked to develop new campus management software for your college. How will you apply Means-Ends Analysis to develop the software?

CO-1

(6)

b)

Given the area of a circle, write a Python program to calculate the radius of the circle using the math module.

CO-1

(3)

Module -2

11

a)

Write a pseudocode to implement a grading system based on the following conditions.

• Marks ≥ 90: Grade A

• Marks 80–89: Grade B

• Marks 70–79: Grade C

• Marks 60-70: Grade D

• Marks 50-69: Grade E

• Marks < 50 : Grade F

• If the difference between the marks and the next multiple of 10 is less that 3, then convert it to next grade.

• If the marks are below 50, no need to convert even if the difference is less than 3.

CO-2

(6)

b)

You are a software developer working on a data analysis tool for a logistics company. The company has a list of delivery times, where each entry represents the time taken (in minutes) for a particular delivery. The team needs to identify delivery times that are divisible by 7 (as they are considered optimal delivery durations) but not divisible by 3 (as they tend to involve more complexity in scheduling). Write an algorithm to print the delivery times that are divisible by 7 but not divisible by 3 from a given set of delivery times.

(3)

12

a)

You are developing a feature for a business analytics tool that processes a list of sales figures for a company. As part of the analytics, the tool must identify the highest and lowest sales figures from the provided data. Write the pseudocode to determine the largest and smallest numbers in a given list of N numbers, ensuring the solution is efficient for real-world applications.

CO-2

(6)

b)

Write the pseudocode to calculate the factorial of a given number using a recursive function.

CO-2

(3)

Module -3

13

a)

You are working as a software engineer for a financial analysis company. The company uses Fibonacci sequences to model certain financial growth patterns, such as project milestones or revenue projections. You are tasked with developing a tool that generates the first ‘n’ numbers in the Fibonacci sequence to assist in creating projections for clients. You should use recursion to solve it.

CO-3

(5)

b)

Write a program that accepts the length of three sides of a triangle as input and determine whether or not the triangle is a right triangle.

CO-3

(4)

14

a)

You are given two positive integers, a and b. Write a Python program using recursion to find the Greatest Common Divisor (GCD) of these integers. You are not allowed to directly calculate the GCD formula, but must instead use the relationship between GCD and LCM.

CO-3

(6)

Module -4

15

a)

You are working as a financial analyst for a bank. The bank has received a list of loan interest rates from multiple branches, and you need to identify the two highest rates to recommend the most cost-effective options to customers. The rates are provided as an array of positive integers, and your task is to develop an efficient algorithm using the Divide and Conquer approach to find the sum of the two largest rates. Explain with a suitable example.

CO-4

(6)

b)

Compare and contrast greedy approach and dynamic programming approach.

CO-4

(3)

16

a)

Explain memoization and tabulation techniques in dynamic programming for calculating the n-th Fibonacci number

CO-4

(6)

b)

Solve the following:

You are given an array of positive integers where each integer represents the time taken to complete a task. Develop a greedy algorithm to determine the maximum number of tasks you can complete in a given total time limit.

Explain your reasoning behind each step of the solution.

CO-4

(3)

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CO-1

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CO-3

(5)

b)

Write a program